Object Oriented Programming Lab (CSEG1044) Experiment-03&04: Operators + Control Flow (Student Handout — Before Submission)

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1 About this handout

This is a **pre-submission** handout. Do the implementation yourself; this document provides the required tasks and some hints.

2 Experiment 03 (Syllabus)

- **Objective:** To apply data types and operators in Java programs.
- **Problem Statement:** Write a Java program demonstrating the use of primitive data types and operators.
- **Expected Outcome:** Students will understand data manipulation using operators.

3 Experiment 04 (Syllabus)

- **Objective:** To implement control flow using conditional and looping statements.

- **Problem Statement:** Design a Java program using decision-making and looping constructs.
- **Expected Outcome:** Students will be able to control program flow effectively.

4 What you must do (minimum requirements)

4.1 Task A — Types + operators mini-lab (Exp-03)

Create `TypesOperatorsDemo.java` that:

1. reads two integers `a` and `b`,
2. prints `a+b`, `a-b`, `a*b`,
3. if `b != 0`, prints:
   - integer division: `a / b`
   - floating division: `a / (double) b`
   - remainder: `a % b`
4. demonstrates precedence:
   - `2 + 3 * 4`
   - `(2 + 3) * 4`
5. demonstrates casting:
   - widening: `int -> double`
   - narrowing: `double -> int` (show truncation)
6. prints one boolean expression using `&&` and `||`.

4.2 Task B — Menu-driven utility program (Exp-04)

Create `MenuUtility.java` that repeats until Exit:

1. Even/Odd check
2. Factorial (loop)
3. Prime check
4. Exit

Requirements:

- Use `switch` for menu choice handling.
- Use loops for factorial and prime check.
- Handle invalid menu choice.

- Show at least one `continue` usage (example: skip processing if input is negative).

5 Hints (read only if stuck)

- `5/2` is integer division (result `2`); use `5/2.0` for `2.5`.
- Always check `b==0` before division/modulo.
- For prime check, loop only up to `i*i <= n`.
- In `switch`, forgetting `break` causes fall-through.

6 What to submit (MANDATORY)

Submit a single zipped folder containing:

- `TypesOperatorsDemo.java`
- `MenuUtility.java`
- sample output for both programs
- observations (5–10 lines): integer vs floating division; where loops are used; common mistakes you faced

Sources: Schildt, *Java: A Beginner's Guide* (10e), McGraw-Hill (Java fundamentals + OOP basics).; Oracle Java Tutorials: Getting Started + Language Basics + Classes and Objects (accessed March 3, 2026).; Java SE API docs (e.g., `String`, `Scanner`) (accessed March 3, 2026).